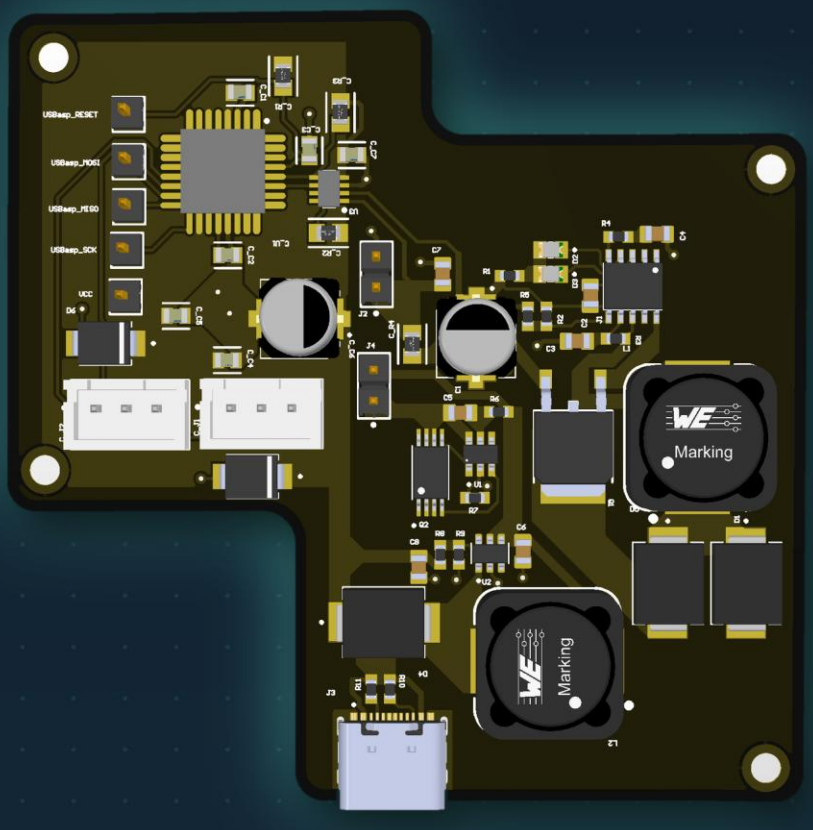
PRELIMINARY DATASHEET

STC-01

Solar Tracker Power and Control Board

MPPT solar charging, Li-ion protection, a regulated 5 V rail and I²C power monitoring on one two-layer board

20 W	2.0 A	5.0 V	3.0 A
PV array, 4 × 5 W	Charge current	Regulated output	Protection trip (typ.)



PRELIMINARY DATASHEET

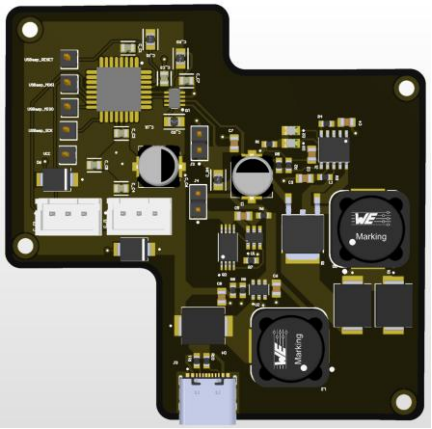
STC-01

Solar Tracker Power and Control Board

MPPT solar charger · Li-ion protection · 5 V boost · ATmega328P · I²C power monitor

GENERAL DESCRIPTION

The STC-01 is a compact two-layer board that powers and controls a small dual-axis solar tracker. It harvests energy from a photovoltaic (PV) array with a maximum power point tracking (MPPT) charger, stores it in a single-cell Li-ion battery behind a hardware protection circuit, and generates a regulated 5 V rail for the ATmega328P controller and the servo outputs. An INA219 power monitor on the I²C bus reports voltage and current to the controller.



Top view of the STC-01 (3D render)

FEATURES

- MPPT solar charger (CN3791), 4.2 V ±1 %, charge current set to 2.0 A
- PV input set-point 9 V nominal (4 × 9 V / 5 W panels in parallel, 20 W nominal)
- Dual-MOSFET battery protection (DW01A + FS8205A): over-voltage, under-voltage, over-current and short-circuit
- 5.0 V boost output (MT3608), 1 A continuous design load
- ATmega328P controller with two 3-pin servo outputs
- INA219 PV power monitor: 10 mΩ shunt, 1 mA current step, I²C
- Servo supply with 470 µF bulk capacitor and a TVS clamp on each output
- USB Type-C 5 V input and 5-pin USBasp programming header
- Two-layer layout verified against the NTI fabrication rules (0.25 mm track, 0.3 mm clearance)

APPLICATIONS

- Dual-axis solar trackers and off-grid sensor nodes
- Educational and research platforms for MPPT and battery management

Key parameter	Value
PV input	4.5 to 12 V recommended open-circuit voltage; 9 V nominal MPPT set-point
Battery	1S Li-ion, 4.2 V charge voltage, 2.0 A maximum charge current
Protection trip points	4.30 V over-voltage, 2.40 V under-voltage, 3.0 A over-current (typ.)
Output	5.0 V, 1 A continuous (design load)
Controller	ATmega328P, SPI programming (USBasp), I²C bus
Monitoring	INA219 on the PV path: 10 mΩ shunt, 1 mA current step, 4 mV voltage step
Operating temperature	0 to +45 °C (limited by Li-ion charging)

1. Functional overview

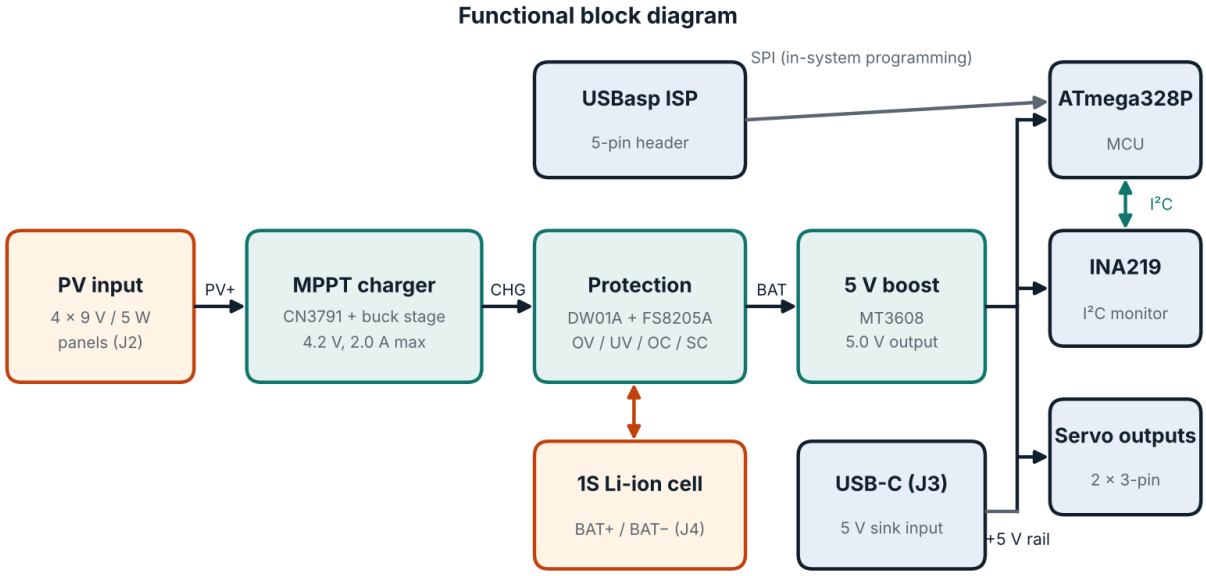


Figure 1. Functional block diagram.

1.1 MPPT charger

A CN3791 buck charger regulates the PV input voltage to a programmable maximum power point set by a resistor divider, so the array operates near its maximum power voltage. The charger runs in three phases: trickle charge below 66.5 % of the float voltage, constant current at the programmed limit, and constant voltage at 4.2 V. A new cycle starts when the cell falls below 95.5 % of the float voltage.

1.2 Battery protection

A DW01A controller drives two N-channel MOSFETs in an FS8205A dual package placed in the negative battery path. The controller opens the discharge switch on under-voltage or over-current, and opens the charge switch on over-voltage. The battery negative terminal (BAT−) is separate from system ground and joins it only through the two MOSFETs.

1.3 5 V boost converter

An MT3608 current-mode boost converter raises the battery voltage (3.0 to 4.2 V) to a fixed 5.0 V set by the feedback divider. The 5 V rail supplies the microcontroller, the INA219 and the servo outputs.

1.4 Controller and monitoring

The ATmega328P is programmed through the 5-pin USBasp header (RESET, MOSI, MISO, SCK, VCC). It reads the INA219 over I²C to obtain the PV voltage and current (the 10 mΩ shunt sits in the PV input path, between SOLAR_OUT and BUCK_IN). This is the measurement used by an angle-based perturb-and-observe tracking algorithm. The battery current is not measured.

2. Board overview

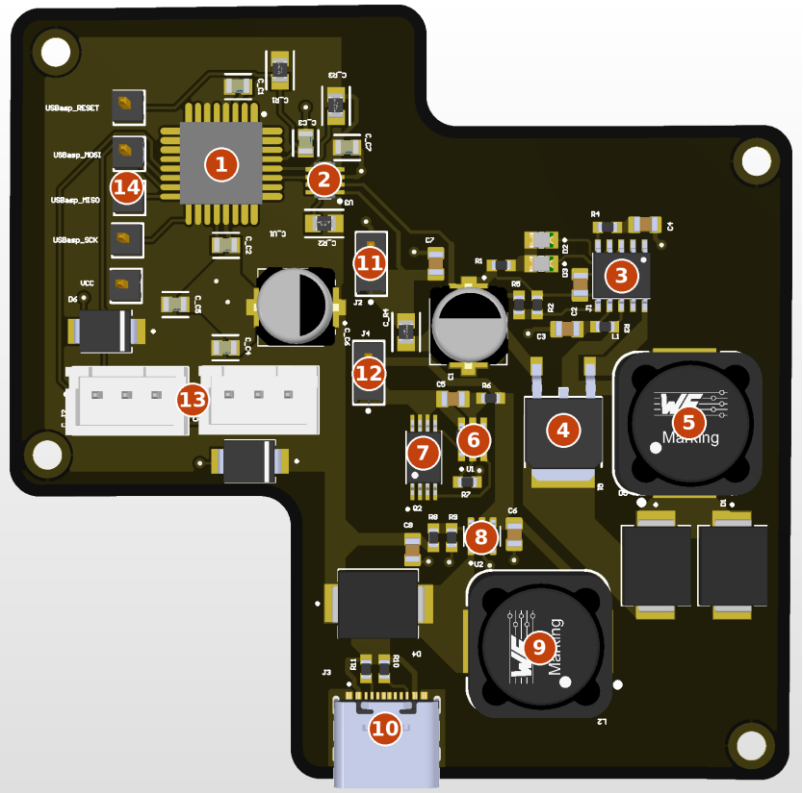


Figure 2. Component locations (top side).

#	Item	Function
1	ATmega328P-AU (TQFP-32)	Controller, servo PWM, I²C master
2	INA219 (A grade)	I²C PV voltage and current monitor
3	CN3791	MPPT Li-ion charge controller
4	P-channel MOSFET	Buck switch of the charger
5	Charger inductor	Buck inductor of the charger
6	DW01A	Battery protection controller
7	FS8205A	Dual N-MOSFET protection switch
8	MT3608	5 V boost converter
9	Boost inductor	Boost converter inductor
10	J3, USB Type-C	5 V power input
11	J2	PV input header
12	J4	Battery header
13	C_J1, C_J2	Servo output connectors (3-pin)
14	USBasp header	In-system programming

3. Specifications

Absolute maximum ratings are device-level limits from the component datasheets. Stresses above these ratings may cause permanent damage. Board-level limits are set by the lowest-rated capacitor and diode on each net.

3.1 Absolute maximum ratings

Parameter	Rating	Limited by
PV input voltage (device level)	30 V	CN3791
Boost input voltage (device level)	24 V	MT3608
INA219 bus voltage	26 V	INA219
Microcontroller supply voltage	6.0 V	ATmega328P
Storage temperature	−40 to +125 °C	DW01A family

3.2 Recommended operating conditions

Parameter	Conditions	Min	Typ	Max	Unit
PV open-circuit voltage	Array in parallel	4.5	10.5	12	V
PV short-circuit current	4 × 0.6 A		2.4		A
Battery voltage	1S Li-ion	3.0	3.7	4.2	V
5 V load current	Continuous	0		1.0	A
Ambient temperature	Battery charging	0		45	°C

3.3 Electrical characteristics

Unless noted, T_A = 25 °C and the cell voltage is 3.7 V. “IC datasheet” values are quoted from the component manufacturer. “Calculated” values are derived from the component values on this board and have not been verified by measurement.

Parameter	Conditions	Min	Typ	Max	Unit	Basis
MPPT charger (CN3791)						
PV input voltage range	IC operating range	4.5		28	V	IC datasheet
MPPT pin regulation voltage		1.18	1.205	1.23	V	IC datasheet
PV regulation set-point	R_top/R_bot = 150 kΩ / 23.2 kΩ	8.8	9.0	9.2	V	Calculated
Battery regulation voltage		4.158	4.20	4.247	V	IC datasheet
Charge current limit	R_cs = 60 mΩ	1.83	2.00	2.17	A	Calculated
Trickle-charge threshold	64 to 69 % of float	2.69	2.79	2.90	V	Calculated
Trickle-charge current	17.5 % of limit		0.35		A	Calculated
Recharge threshold	95.5 % of float		4.01		V	Calculated
Charge termination current	16 % of limit		0.32		A	Calculated
Battery protection (DW01A + FS8205A)						
Over-voltage protection		4.25	4.30	4.35	V	IC datasheet
Over-voltage release		4.05	4.10	4.15	V	IC datasheet
Under-voltage protection		2.30	2.40	2.50	V	IC datasheet
Under-voltage release		2.90	3.00	3.10	V	IC datasheet
Discharge over-current detect	Across both FETs	120	150	180	mV	IC datasheet
Discharge over-current trip	R_ds(on) total 50 mΩ	2.4	3.0	3.6	A	Calculated
Short-circuit detect			1.35		V	IC datasheet
Short-circuit trip current	R_ds(on) total 50 mΩ		27		A	Calculated
Over-voltage delay		100		200	ms	IC datasheet
Under-voltage delay		50		100	ms	IC datasheet

Parameter	Conditions	Min	Typ	Max	Unit	Basis
Over-current delay		10		20	ms	IC datasheet
MOSFET loss at 2 A	$I^2 \times 50\ m\Omega$		0.20		W	Calculated
5 V boost (MT3608)						
Input voltage range	IC operating range	2		24	V	IC datasheet
Battery input range	Protected cell	3.0	3.7	4.2	V	Design
Output voltage	R_top/R_bot = 110 kΩ / 15 kΩ		5.0		V	Calculated
Switching frequency			1.2		MHz	IC datasheet
Switch current limit			4		A	IC datasheet
Continuous output current	Design load			1.0	A	Design
Battery current at 1 A out	η = 85 %, V_bat 4.2 to 3.0 V	1.40	1.59	1.96	A	Calculated
Power monitor (INA219)						
Shunt resistor	PV input path		10		mΩ	Design
Shunt voltage step	All PGA ranges		10		µV	IC datasheet
Current step	10 µV / 10 mΩ		1.0		mA	Calculated
Full-scale current	PGA ÷1, ±40 mV		±4.0		A	Calculated
Shunt offset voltage	A grade, PGA ÷1			±100	µV	IC datasheet
Equivalent current offset	100 µV / 10 mΩ			±10	mA	Calculated
Bus voltage step			4		mV	IC datasheet
Bus voltage range		0		26	V	IC datasheet
Conversion time	12-bit, no averaging		532		µs	IC datasheet
Supply voltage		3.0		5.5	V	IC datasheet
Supply current			0.7	1.0	mA	IC datasheet

4. Interfaces

Ref.	Type	Signals and function
J2	2-pin header	PV+ and GND: solar array input
J4	2-pin header	BAT+ and BAT−: single-cell Li-ion battery
J3	USB Type-C receptacle	5 V power input; CC1 and CC2 pulled down for sink operation (R10, R11)
C_J1, C_J2	3-pin connector, 2.54 mm	Servo outputs: pin 1 PWM (OC1A / OC1B), pin 2 +5 V, pin 3 GND. Each output has a 100 nF capacitor and a TVS diode; 470 µF shared bulk capacitor
USBasp	5 single-pin headers	RESET, MOSI, MISO, SCK, VCC. The VCC pin connects to the +5 V rail: do not power the board from the programmer and another source at the same time

4.1 I²C bus

The INA219 is an I²C slave (seven-bit address range 0x40 to 0x4F, selected by its A0 and A1 pins) and supports clock rates up to 400 kHz in fast mode. The ATmega328P is the bus master (SDA on PC4, SCL on PC5) with 10 kΩ pull-up resistors to +5 V. The shunt voltage register has a fixed step of 10 µV and the bus voltage register a step of 4 mV, so the current step is 10 µV divided by the shunt resistance, which is 1 mA for the 10 mΩ shunt.

4.2 Microcontroller

The ATmega328P-AU (32-pin TQFP) runs from the +5 V rail with 100 nF decoupling on VCC and AREF. No crystal is fitted, so the device uses its internal RC oscillator. The two servo signals use the hardware PWM outputs OC1A (PB1) and OC1B (PB2). SPI programming uses PB3 to PB5, and RESET has a 10 kΩ pull-up and a 100 nF capacitor. The ADC inputs are not used.

5. Application information

5.1 Setting the PV operating point

The charger regulates the PV voltage to V_MPPT, set by the divider on the MPPT pin:

```
V_MPPT = 1.205 V × (1 + R_top / R_bot)
```

For a 9 V nominal maximum power point, R_top = 150 kΩ and R_bot = 23.2 kΩ give 9.0 V. Choose the set-point near the V_mp of the panel at the expected cell temperature.

5.2 Setting the charge current

```
I_CHG = 0.12 V / R_cs        R_cs = 60 mΩ  →  I_CHG = 2.0 A
```

Connect the CSP and BAT pins directly to the two terminals of the sense resistor (Kelvin connection) to preserve accuracy.

5.3 Setting the 5 V output

```
V_OUT = 0.6 V × (1 + R_top / R_bot)   110 kΩ / 15 kΩ  →  5.0 V
I_BAT = 5 V × I_OUT / (η × V_BAT) = 1.96 A   (η = 0.85, 1 A, 3.0 V)
```

5.4 Protection trip currents

```
I_OC = V_OI1 / (2 × R_ds(on)) = 150 mV / 50 mΩ = 3.0 A   (2.4 to 3.6 A)
I_SC = V_OI2 / (2 × R_ds(on)) = 1.35 V / 50 mΩ ≈ 27 A
```

The charge current of 2.0 A produces 100 mV across the two MOSFETs, which stays below the charge-direction detection threshold of the protection controller. The sense voltage is a function of the on-resistance of the MOSFETs, so it rises with temperature.

5.5 Layout notes

- Keep the input capacitors, switch and diode of each converter in a tight loop; route no signal under the switching nodes.
- Power nets (battery, PV and 5 V) use 1.5 mm tracks; the minimum track and clearance are 0.25 mm and 0.3 mm.
- Keep BAT− separate from system ground; the two connect only through the protection MOSFETs.

- Keep the battery away from the inductors and diodes; do not charge Li-ion cells above 45 °C.

6. Mechanical data

Item	Description
Board	Two-layer, L-shaped outline
Approximate size	70 × 68 mm
Mounting holes	4
Minimum track / clearance	0.25 mm / 0.3 mm
Minimum polygon pour clearance	0.4 mm
Via	0.9 mm pad, 0.4 mm hole
Fabrication rule set	NTI

7. Revision history

Revision	Date	Description
0.1	October 2026	Preliminary release

Preliminary information: specifications may change without notice. Values marked “Calculated” are design values and are not guaranteed.